WORKED SOLUTION

STEP III 2003, Question 2

Pure • Binomial Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q2

The key idea

The central binomial coefficient $\binom{2r}{r}$ hides inside the binomial series for negative half-integer powers. Once the opening identity splits $(2r)!$ into its odd and even factors, the coefficient of p^r in $(1-p)^{-1/2}$ turns out to be exactly $\binom{2r}{r}/4^r$. Each sum is then a binomial series in disguise, and the only job is to pick p (and, in (ii), the right power).

Solution

The identity

Split the factors of $(2r)!$ into odd and even ones:

$$(2r)! = [1 \times 3 \times \cdots \times (2r-1)] \times [2 \times 4 \times \cdots \times 2r].$$

Each of the r even factors is 2 times one of $1, 2, \ldots, r$, so the second bracket is $2^r\, r!$. Hence

$$\binom{2r}{r} = \frac{(2r)!}{r!\, r!} = \frac{1 \times 3 \times \cdots \times (2r-1) \times 2^r\, r!}{r!\, r!} = \frac{1 \times 3 \times \cdots \times (2r-1)}{r!} \times 2^r,$$

as required.

Part (i)

For $|p| < 1$, the binomial series gives

$$\begin{aligned}(1-p)^{-\frac{1}{2}} &= 1 + \left(-\tfrac{1}{2}\right)(-p) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2!}(-p)^2 \\ &\quad + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{3!}(-p)^3 + \cdots \\ &= 1 + \tfrac{1}{2}p + \tfrac{3}{8}p^2 + \tfrac{5}{16}p^3 + \cdots.\end{aligned}$$

The general term follows the same pattern. In the coefficient of p^r there are r minus signs from the factors $-\frac{1}{2}, -\frac{3}{2}, \ldots$ and r more from $(-p)^r$, so they cancel, leaving

$$\frac{\frac{1}{2} \times \frac{3}{2} \times \cdots \times \frac{2r-1}{2}}{r!}\, p^r = \frac{1 \times 3 \times \cdots \times (2r-1)}{2^r\, r!}\, p^r.$$

Ask yourself How does this connect to $\binom{2r}{r}$?

By the identity, $\dfrac{1 \times 3 \times \cdots \times (2r-1)}{r!} = \dfrac{\binom{2r}{r}}{2^r}$. So the coefficient of p^r is $\binom{2r}{r}/4^r$, which also holds for $r = 0$ since both sides are 1. Therefore

$$(1-p)^{-\frac{1}{2}} = \sum_{r=0}^{\infty} \binom{2r}{r} \left(\frac{p}{4}\right)^r \qquad (|p| < 1).$$

The sum we want has $(p/4)^r = 1/8^r$, so we take $p = \frac{1}{2}$, which is inside the interval of convergence. Then

$$\sum_{r=0}^{\infty} \frac{\binom{2r}{r}}{8^r} = \left(1 - \tfrac{1}{2}\right)^{-\frac{1}{2}} = \left(\tfrac{1}{2}\right)^{-\frac{1}{2}} = \sqrt{2}.$$

Part (ii)

The extra factor $2r+1$ suggests going one power further, to $(1-p)^{-3/2}$. For $|p| < 1$, the coefficient of p^r in its binomial series is, with the signs cancelling as before,

$$\begin{aligned}\frac{\frac{3}{2} \times \frac{5}{2} \times \cdots \times \frac{2r+1}{2}}{r!} &= \frac{3 \times 5 \times \cdots \times (2r+1)}{2^r\, r!} \\ &= (2r+1)\, \frac{1 \times 3 \times \cdots \times (2r-1)}{2^r\, r!} = \frac{(2r+1)\binom{2r}{r}}{4^r},\end{aligned}$$

since $3 \times 5 \times \cdots \times (2r+1) = (2r+1) \times [1 \times 3 \times \cdots \times (2r-1)]$, and then using the identity exactly as in (i). (For $r = 0$ both ends equal 1.) So

$$(1-p)^{-\frac{3}{2}} = \sum_{r=0}^{\infty} (2r+1) \binom{2r}{r} \left(\frac{p}{4}\right)^r \qquad (|p| < 1).$$

Now we need $p/4 = \frac{1}{5}$, that is $p = \frac{4}{5}$, again with $|p| < 1$. Hence

$$\sum_{r=0}^{\infty} \frac{(2r+1)\binom{2r}{r}}{5^r} = \left(1 - \tfrac{4}{5}\right)^{-\frac{3}{2}} = \left(\tfrac{1}{5}\right)^{-\frac{3}{2}} = 5^{\frac{3}{2}} = (\sqrt{5})^3,$$

as required.

Alternatively, write $f(x) = \sum \binom{2r}{r} x^r = (1-4x)^{-1/2}$ for $|x| < \frac{1}{4}$. Differentiating the power series term by term inside its interval of convergence, $\sum r\binom{2r}{r} x^r = x f'(x) = 2x(1-4x)^{-3/2}$, so

$$\sum (2r+1) \binom{2r}{r} x^r = 4x(1-4x)^{-\frac{3}{2}} + (1-4x)^{-\frac{1}{2}} = (1-4x)^{-\frac{3}{2}},$$

and $x = \frac{1}{5}$ gives the same answer.

Checks

- The first few terms of (i) are $1+\frac{2}{8}+\frac{6}{64}+\frac{20}{512}\approx 1.371$, already close to $\sqrt{2}\approx 1.414$; summing 200 terms gives $1.41421356\ldots$.
- In (ii) the series converges more slowly (ratio of terms tends to $\frac{4}{5}$), but 400 terms give $11.18033988\ldots$, matching $5\sqrt{5}\approx 11.1803$.
- The coefficients $\binom{2r}{r}/4^r$ for $r=0,1,2,3$ are $1, \frac{1}{2}, \frac{3}{8}, \frac{5}{16}$, agreeing with the four terms found directly.

Common slips

- Losing track of signs: the r negative factors in the coefficient and the $(-1)^r$ from $(-p)^r$ always cancel, so every coefficient is positive.
- Forgetting to check that the chosen p satisfies $|p|<1$; without it, the series need not converge to the function.
- In (ii), using $(1-p)^{-1/2}$ again with a different p: the factor $2r+1$ cannot come from that series alone.

Beyond STEP

The function $\sum\binom{2r}{r}x^r=(1-4x)^{-1/2}$ is the generating function of the central binomial coefficients, a standard tool in combinatorics; for example, squaring it gives $\sum_r 4^r x^r$, which proves $\sum_k \binom{2k}{k}\binom{2r-2k}{r-k}=4^r$. Justifying term-by-term differentiation, and knowing exactly where a power series converges, are first-year analysis topics. See Abbott, *Understanding Analysis*, for power series, and Siklos, *Advanced Problems in Mathematics*, for more STEP practice with series.

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